

Sarthak Consul

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Education

Indian Institute of Technology Bombay, Mumbai, India
Bachelor of Technology (with Honours) in Electrical Engineering

July 2016 - Present

- **Overall CPI:** 9.63/10
- **Minor** in Computer Science and Engineering

Publications

Analysis of Lower Bounds for Simple Policy Iteration (preprint)
*Sarthak Consul**, Bhishma Dedhia*, Kumar Ashutosh*, Parthasarathi Khirwadkar*
{arXiv:1911.12842} *Equal Contribution

Research Internships & Projects

Singing-Style Transfer across Unpaired Music Clips

IIT Bombay

Guide: *Prof. Subhasis Chaudhuri*, Department of Electrical Engineering

Aug 2019 - Present

- Working on using a **WGAN** to transform the vocals of the song to mimic the singer of an unpaired target song
- Developing a perceptual loss that separates rhythmic and lyrical similarities from timbral similarities
- Analyzing the features extracted by a Conditional Variational Autoencoder (**CVAE**) and various classifiers

Leveraging Reinforcement Learning for Semantic Segmentation

IIT Bombay

Guide: *Prof. Amit Sethi*, Department of Electrical Engineering

Aug 2019 - Present

- Using an **Asynchronous Actor-Critic Network (A3C)** to generate a segmentation mask from the input image
- Trained a sequential labelling agent to achieve a **53.62%** mIoU on the PASCAL VOC 2012 dataset
- Devised a novel hierarchical segmentation approach which is currently being analyzed and tested

Analysis of Lower Bounds for Simple Policy Iteration

IIT Bombay

Guide: *Prof. Shivaram Kalyanakrishnan*, Department of Computer Science & Engineering

Fall 2019

- Devised a family of n -state, k -action MDPs to obtain a lower bound of $\mathcal{O}((3+k)2^{n/2-3})$ for simple PI
- Theoretically proved the lower bound on the number of iterations required by SPI for any general n and k

Segmentation of Lacunar Objects from Ultra High Resolution μ CT Bone Scans

ETH Zürich

Guide: *Prof. Dr. Ralph Müller*, D-HEST, Institute for Biomechanics

Summer 2019

- Worked on extracting lacunar structures from trabecular bone samples of rare osteoporotic patients
- Modified Fully Convolutional Networks like WNet for segmentation in 3D images in an unsupervised setting
- Worked on a 2 stage method that extracts regions using adaptive thresholding followed by a classifier to fine-tune results by removing noise structures, achieving an F1 score of **98.4%**

Semantic Segmentation of Medical Images using Fully Convolutional Networks

IIT Bombay

Guide: *Prof. Amit Sethi*, EE, Medical Deep learning and AI Lab

Summer 2018

- Implemented Global Convolution on a fully-connected ResNet backbone using PyTorch for the **segmentation of lungs** from chest x-rays and **nuclei segmentation** from h/he stained histopathological sections of organs
- Trained VGG-UNet with watershed post-processing to improve the state of the art results of **nuclei segmentation** to a mean F1 score of **85%** and set the baseline for the **MoNuSeg Challenge** at MICCAI 2018
- Annotated and pre-processed data from ~ 800 patients afflicted with precancerous/cancerous oral lesions

Scholastic Achievements

- Achieved **Advanced Performer** grade (AP) in **4 courses**: *Biology* (15 out of 445), *Introduction to Electrical Systems* (2 out of 149), *Network Theory* (10 out of 143), and *Electronic Devices Lab* (3 out of 139)
- Secured All India Rank 840 (out of around 200,000 candidates) in JEE Advanced [2016]
- Awarded KVPY Fellowship (**AIR 16**) by Department of Science and Technology, Govt. of India [2014]
- Placed in **National Top 1%** in National Standard Examination in Junior Sciences [2012]
- Ranked **4th** in State Science Talent Search Examination (SSTSE), conducted by Govt. of Rajasthan [2014]
- Recipient of NTSE Scholarship by National Council of Educational Research and Training [2012]

Miscellaneous Projects

Adversarial Attacks on Neural Networks

EE782: Advanced Topics in Machine Learning

Prof. Amit Sethi, EE, IIT Bombay

Fall 2019

- Crafted **poison examples** for models trained using transfer learning. Combined this with watermarking to attack models in an end-to-end training scenario, achieving a **96%** poisoning success rate on CIFAR10
- Generated adversarial perturbations using **FGSM** and **Projected Gradient Descent** algorithms
- Compared the performance of membership and gradient queries in mimicking networks in exploratory attacks

Digitally Programmable Analog Computer

Prof. Mukul Chandorkar, EE, IIT Bombay

EE344: Electronics Design Lab

Spring 2019

Designed and developed a standalone digitally programmable analog computer, capable of solving non-linear dynamical systems upto the fifth order, demonstrating hardware-in-loop capabilities. The system is capable of working in real-time, is fully programmable, and has an on-board power management system [Technical Report]

Human Detection from Post-disaster Low-Wavelength IR Imagery

Prof. Vikram M. Gadre, EE, IIT Bombay

EE338: Digital Signal Processing

Spring 2019

- Implemented a system to locate humans from LWIR images for the purpose of streamlining rescue operations
- Extracted regions using **MSER**, followed by usage of local sums, the **Histogram of Oriented Gradients** and the **DCT** coefficients as features for a **SVM** classifier
- Presented to students and teachers of various tier 2 and 3 colleges across the country as part of TEQIP-KIT-2019 workshop under the initiative of the Ministry of Human Resource Development (MHRD), Govt. of India

Audio Source Separation using a Low Latency Neural Network

Prof. Sunita Sarawagi, CSE, IIT Bombay

CS419: Machine Learning

Fall 2018

- Trained a low-latency NN on the DSD100 dataset to separate the different audio sources in a music clip
- Used depthwise separable convolutions, inspired by the GCN architecture to reduce parameters

Tree based Stereo Matching for Postcapture Refocus

Prof. Suyash Awate & Prof. Ajit Rajwade, CSE, IIT Bombay

CS663: Digital Image Processing

Fall 2018

- Implemented a **hybrid tree based** stereo matching algorithm, constructing MSTs to calculate disparity
- Incorporated CLMF (Cross Local Multipoint filter) to use as an edge preserving smoother for refining the depth map, and used the depth estimates to interactively refocus and generate depth blurring

Multicycled and Pipelined Implementation of IITB-RISC ISA

Prof. Virendra Singh, EE, IIT Bombay

EE309: Microprocessors

Fall 2018

- Designed and implemented a 6-stage pipelined multicycle RISC processor in VHDL, consisting of arithmetic, logical and branching instructions, and tested it on Altera DE0-Nano FPGA board
- Implemented Branch Predictors, Priority Encoders and Hazard Detection Units to reduce latencies and eliminate the branch hazards

Analog Sound Locator

Prof. Siddharth Tallur, EE, IIT Bombay

E230: Analog Systems Lab

Spring 2018

- Designed and implemented a compact analog circuit to locate distant sound sources using dynamic phase locking by JFET delay circuits. One of the **winning projects** in the final evaluation (amongst 68 teams)

Mentorship Experience

Department Academic Mentor

2019-Present

- Mentoring 12 second year students with a view to guide and help them cope academically

Teaching Assistant

Prof. Swapneel Mahajan, Department of Mathematics, IIT Bombay

MA207: Partial Differential Equations

Fall 2018

- Taught a class of approximately 60 students. Involved in the setting and evaluation of examination papers

Instructor of Machine Learning Bootcamp

Summer 2018

- Taught the fundamentals of Machine Learning to a class of approximately **130** students
- Explained the basics of ML and covered the theory and implementation of popular algorithms of supervised and unsupervised learning such as **Random Forest Algorithm**, **SVMs**, **CNNs**, and **k-means Clustering**

Technical Skills

Programming Languages

- C, C++, MATLAB, Python, Bash, VHDL, Assembly, Embedded C

ML Libraries

- PyTorch, TensorFlow, Keras

Web Development

- HTML5, CSS3, PHP

Design and Simulation Tools

- EAGLE, AutoCAD, SolidWorks NgSpice, L^AT_EX, Simulink, LabVIEW, Wireshark

Hardware

- Arduino, Intel 8081, Intel 8085, Altera DE0-Nano

Relevant Coursework

Advanced Machine Learning, Digital Image Processing, Intelligent & Learning Agents, Automatic Speech Recognition, Computer Vision, Machine Learning, Computer Networks, Data Structures & Algorithms, Operating Systems, Probability & Random Processes, Digital Signal Processing, Convex Optimization, Matrix Computations

Online Courses

- Convolutional Neural Network for Computer Vision** (Stanford University - CS231n)
- Machine Learning** (Stanford University - CS229 and Coursera)
- Advanced MATLAB for Scientific Computing** (Stanford University - Lagunita) [with Distinction]
- Neural Networks for Machine Learning** (University of Toronto - Coursera)